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COMP 4107 Project Proposal

Background

Multi-agent reinforcement learning is a challenging and active area of research in which multiple agents learn simultaneously within a shared environment. An interesting phenomenon in this area is the emergence of complex strategies from agents that are trained only with simple reward signals. This project will investigate emergent behaviour in a competitive multi-agent setting by training agents to play a territory control game inspired by Paper.io, in which agents compete to claim as much of a grid as possible while attempting to eliminate opponents by cutting their trails. The primary objective of this project is to train competitive agents using deep reinforcement learning and analyze the strategies that emerge during self-play training. A secondary objective is to investigate how the design of the reward function affects the strategies agents learn and their overall performance.

Methods

A custom two-player territory control game will be implemented as a PettingZoo environment to support multi-agent training. The game consists of a fixed-size grid in which each agent moves continuously, leaving a trail behind it. When an agent loops back to its own territory, all cells enclosed by the loop are claimed. If an agent moves onto another agent's active trail, the trailing agent is eliminated and loses its unclaimed territory. Each agent will observe a local view of the grid centred on its position, encoding cell ownership (neutral, self, opponent), trail positions, and the agent's own orientation.

Both agents will be trained simultaneously using Proximal Policy Optimization (PPO) via the stable-baselines3 library, with a convolutional neural network (CNN) processing the local grid observations. Self-play will be used throughout training, meaning each agent always faces the current version of its opponent, creating a co-evolutionary dynamic in which improvements in one agent place pressure on the other to adapt. Hyperparameters such as learning rate, entropy coefficient, and batch size will be allocated through controlled experiments to allow stable training.

Dataset/Environment

Rather than using a pre-existing dataset or benchmark environment, this project will use a custom-built game environment using PettingZoo, a library designed for multi-agent reinforcement learning that provides a standardized interface compatible with stable-baselines3. The game will be implemented as a fixed-size grid (e.g., 20x20 or 30x30 cells), with configurable parameters including grid size, episode length, and observation radius.

Validation & Analysis

Agent performance will be evaluated by measuring win rate, average territory held at the end of episodes, and average survival time across a fixed set of evaluation episodes run periodically during training. Learning curves will be plotted to analyze the stability and convergence of training.

The primary experiment will compare agents trained under different reward structures.These will include a sparse reward that only rewards wins and penalizes deaths, a dense reward that provides incremental feedback at each timestep, and a mixed reward combining territory gain with bonuses for survival-time.

Additionally, each model will finally be tested against a simple rules-based agent to evaluate performance against a baseline.

In addition to quantitative performance metrics, qualitative analysis will be conducted by visualizing agent behaviour at different stages of training. This will allow documentation of how strategies evolve over time, for example from random movement in early training to recognizable tactics such as aggressive expansion loops or deliberate trail-cutting in later training.

Novelty

The majority of multi-agent reinforcement learning research focuses on environments that have been explored numerous times, such as Atari games or simple particle environments. This project uses a custom-built Paper.io-like environment, which has not been studied formally in the literature. The trail-cutting elimination mechanic introduces a strategic layer not present in standard pursuit-evasion or competitive navigation tasks, as agents must simultaneously manage their own vulnerability while attempting to exploit the opponent's. Further, experimenting with reward structure in this novel environment also provides the opportunity to learn more about its impact. The combination of self-play and reward structure experimentation distinguishes this project from standard PPO-based competitive training setups.

Milestones

February 23rd - March 1st

- Core game logic implemented in Pygame (grid, trail tracking, loop closure, elimination)
- Basic rendering of game state (coloured grid cells, agent positions)
- Section 3 of report

March 2nd - March 8th

- PettingZoo environment wrapper completed with observation and action spaces defined
- Rule-based baseline agent implemented for evaluation purposes
- Begin work on section 4 of report

March 9th - March 15th

- Established PPO training code with CNN observation encoder
- Complete section 4 of report

March 16th - March 22nd

- Reward structure experiments
- Evaluation structure complete (win rate, territory owned, survival time)

March 23rd - End

- All training complete, result analysis
- Completion of sections 5 & 6 of report